

The (E)ER model

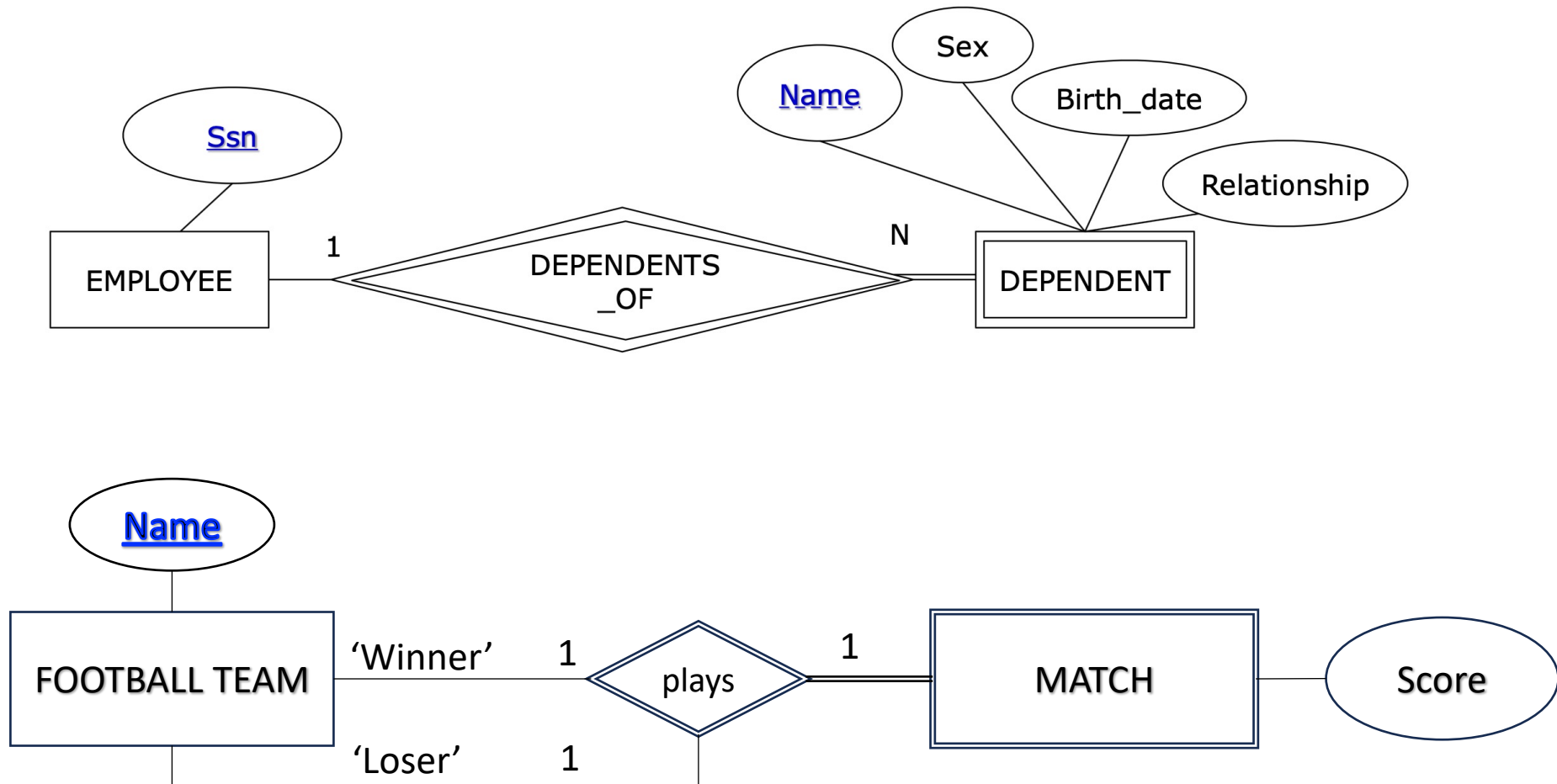
NOTES – Semester 2 – 2023-2024

By: Vo Thi Ngoc Chau

Key attributes for entity types

- One or more key attributes?
- Key attribute:
 - Composite?
 - Multivalued?
 - Derived?
- How about partial key attributes?

Partial keys for weak entity types if there are more than one weak entity in the identifying scope



Neither key nor partial key attributes for
relationship types

Derived attributes can *never* be key/partial key attributes.

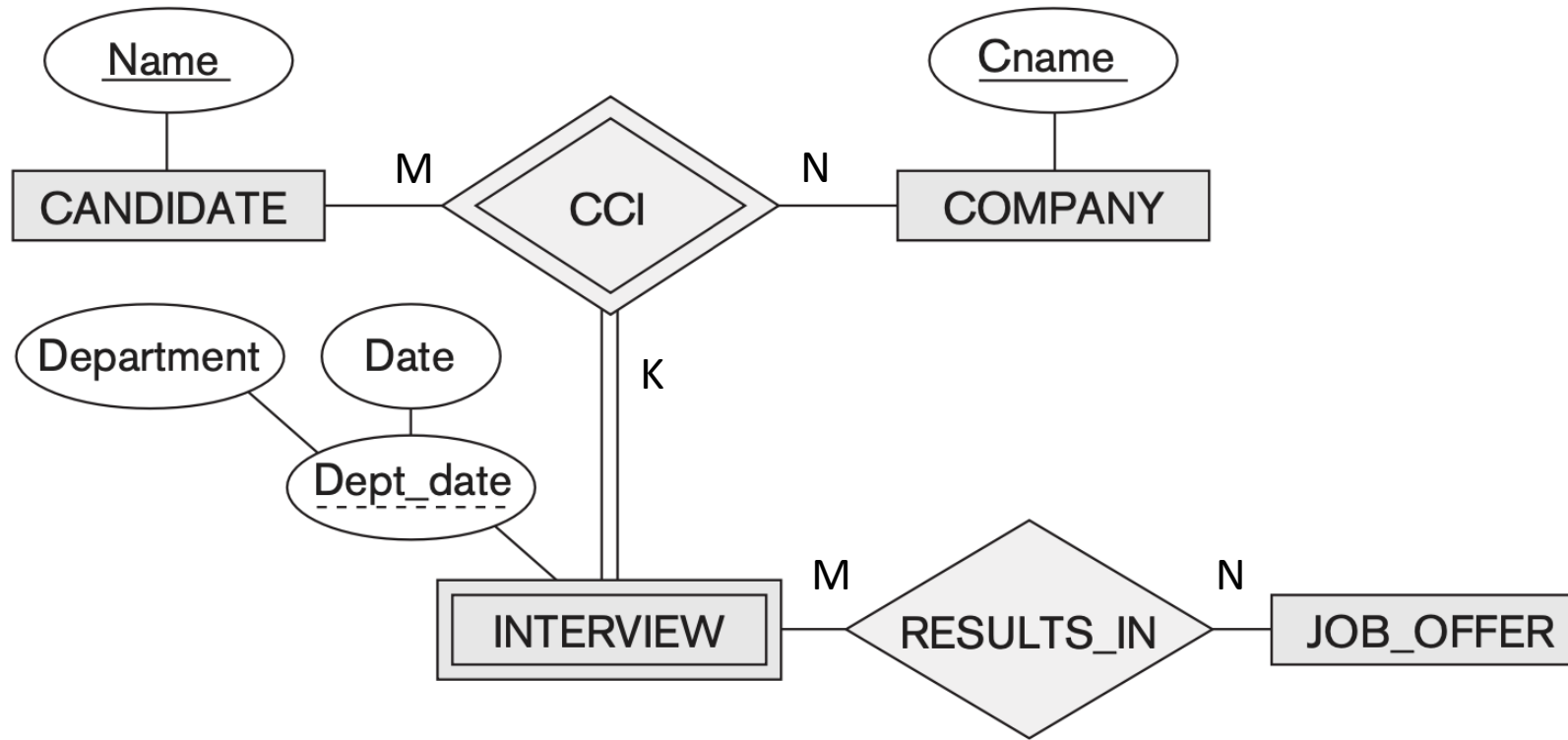
When to change an attribute to entity type

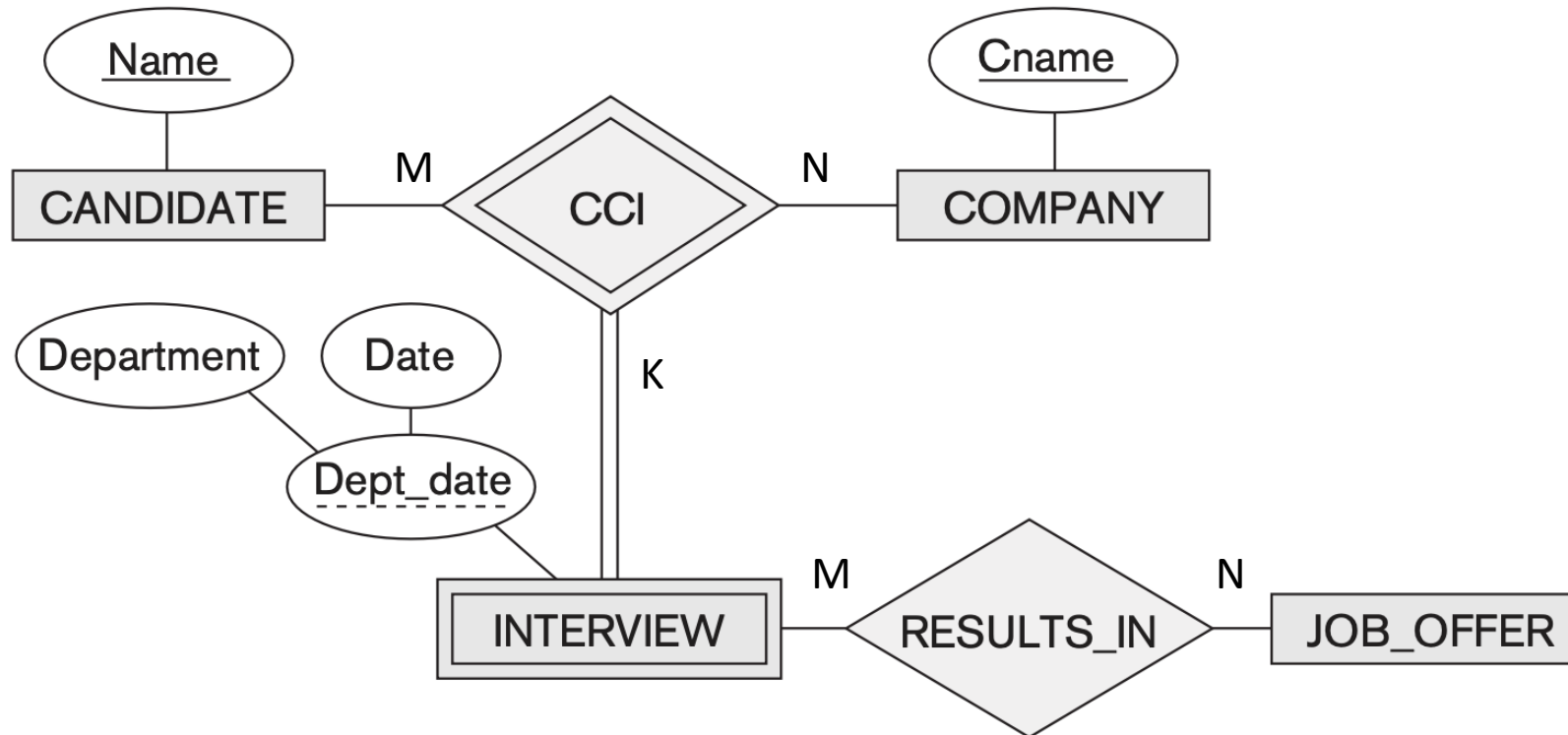
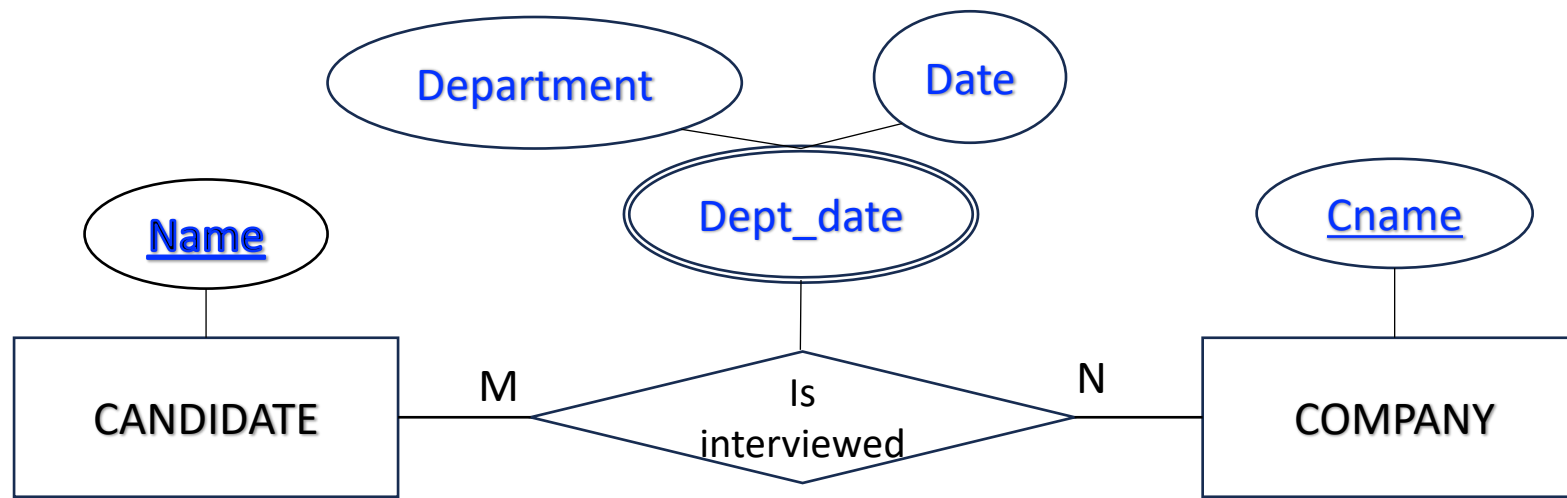
- Model of cars, ships, buildings, ...
- Skills of each employee
- Topics of each presentation
- Authors of books
- Semesters of courses
- ...

When to decide an entity type instead of a relationship type

- Registration: students-courses
- Match: football team-football team
- Order: customers-products
- ...

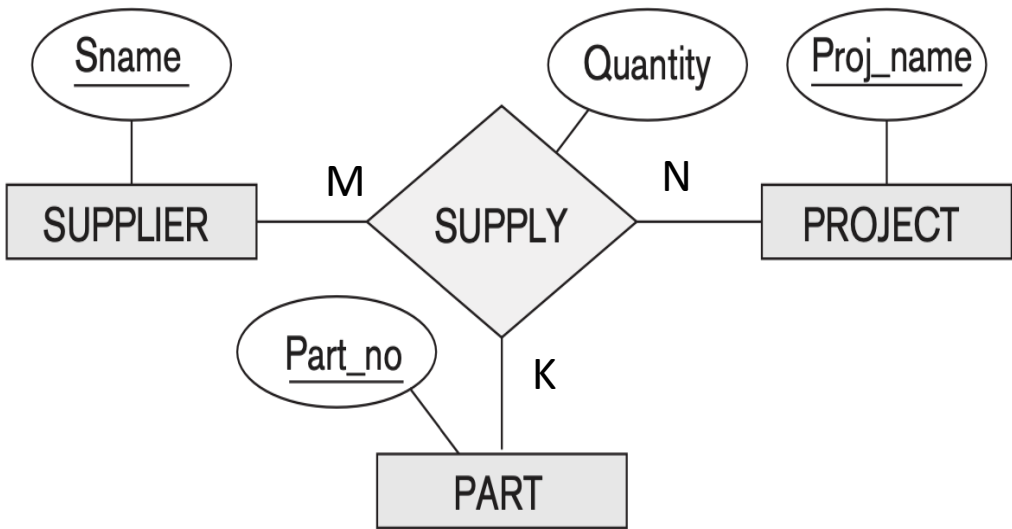
When to decide regular entity types and weak entity types, weak entity types and relationship types



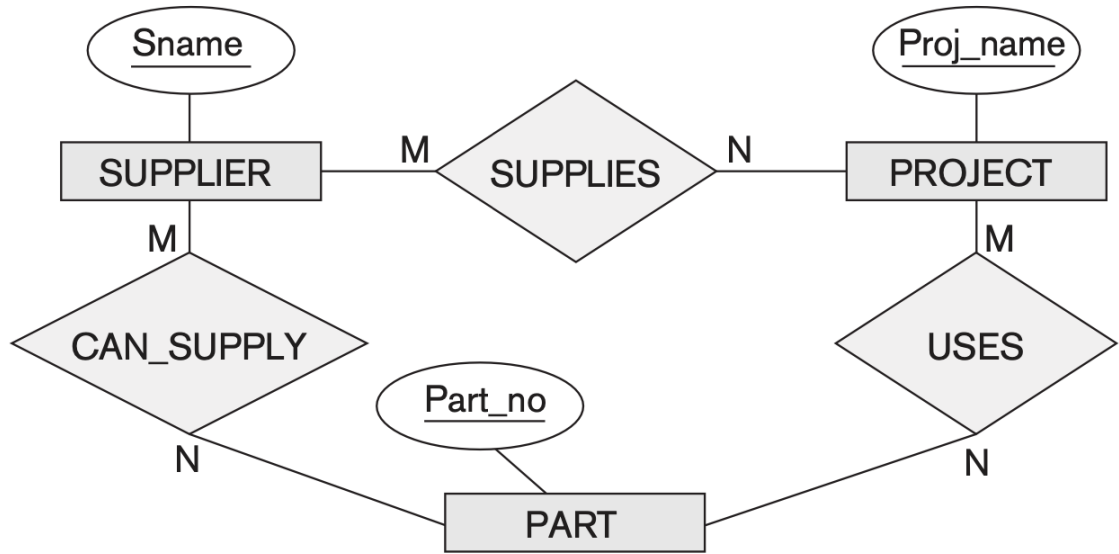


Consider n -ary
relationship types
where $n \geq 3$

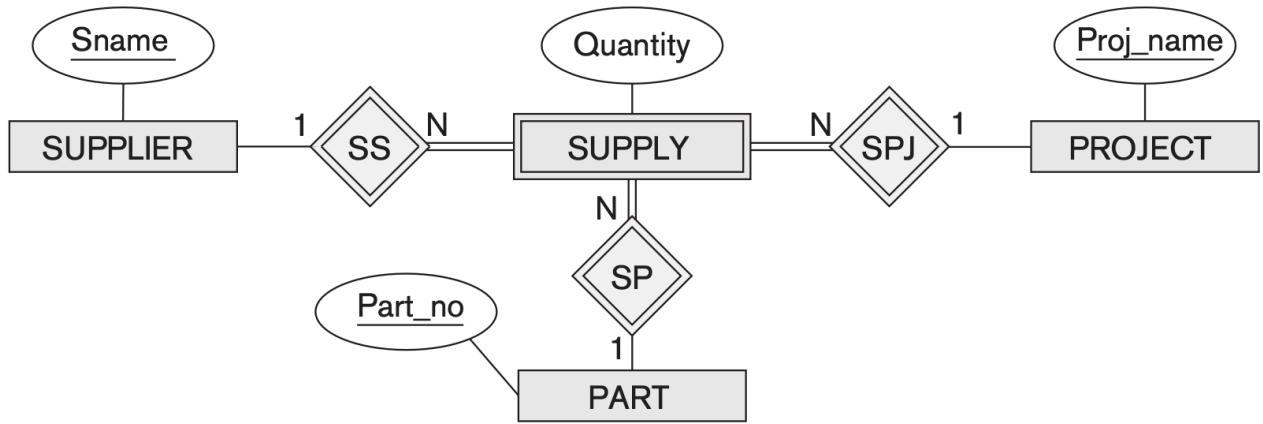
(a)



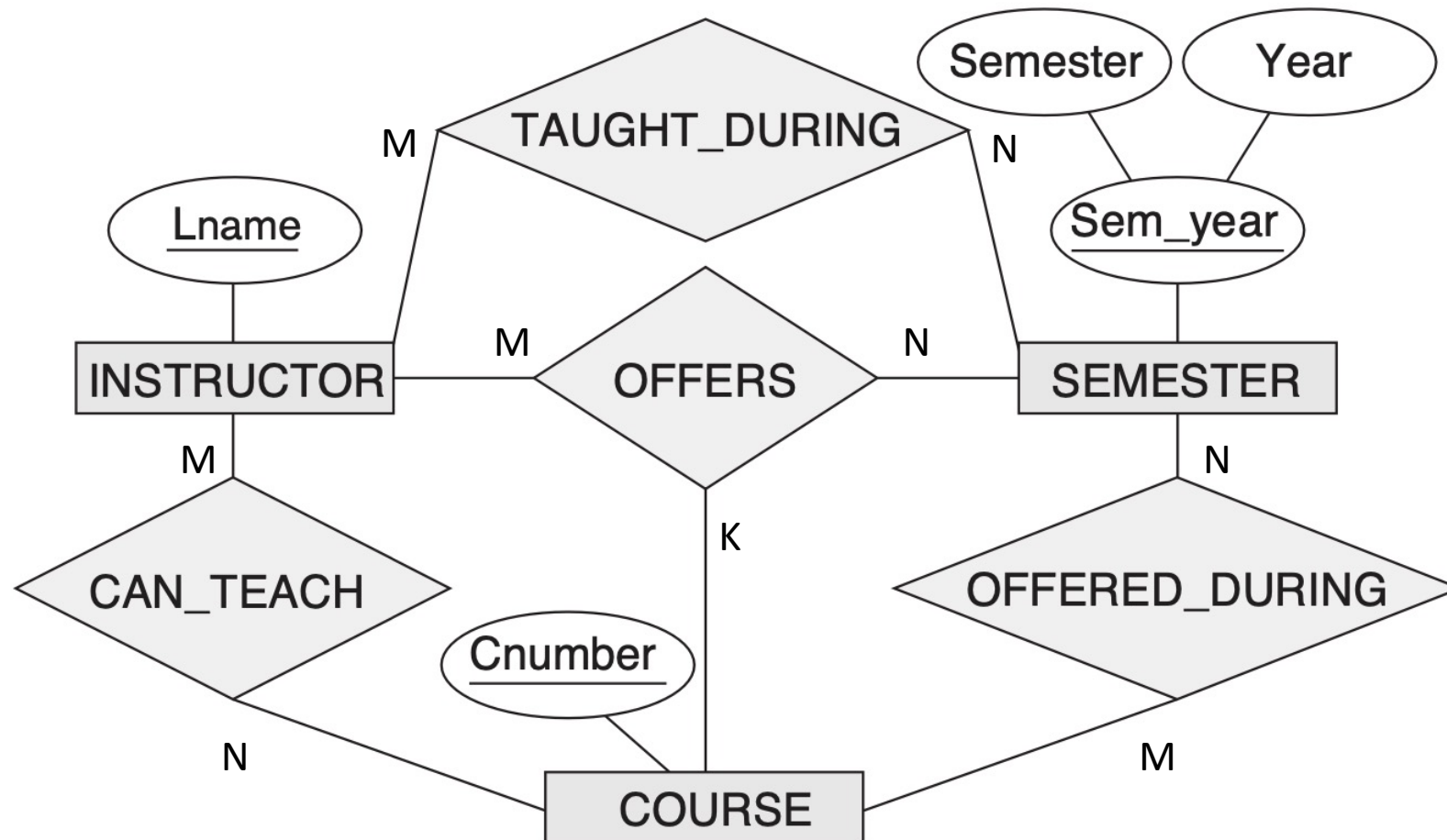
(b)



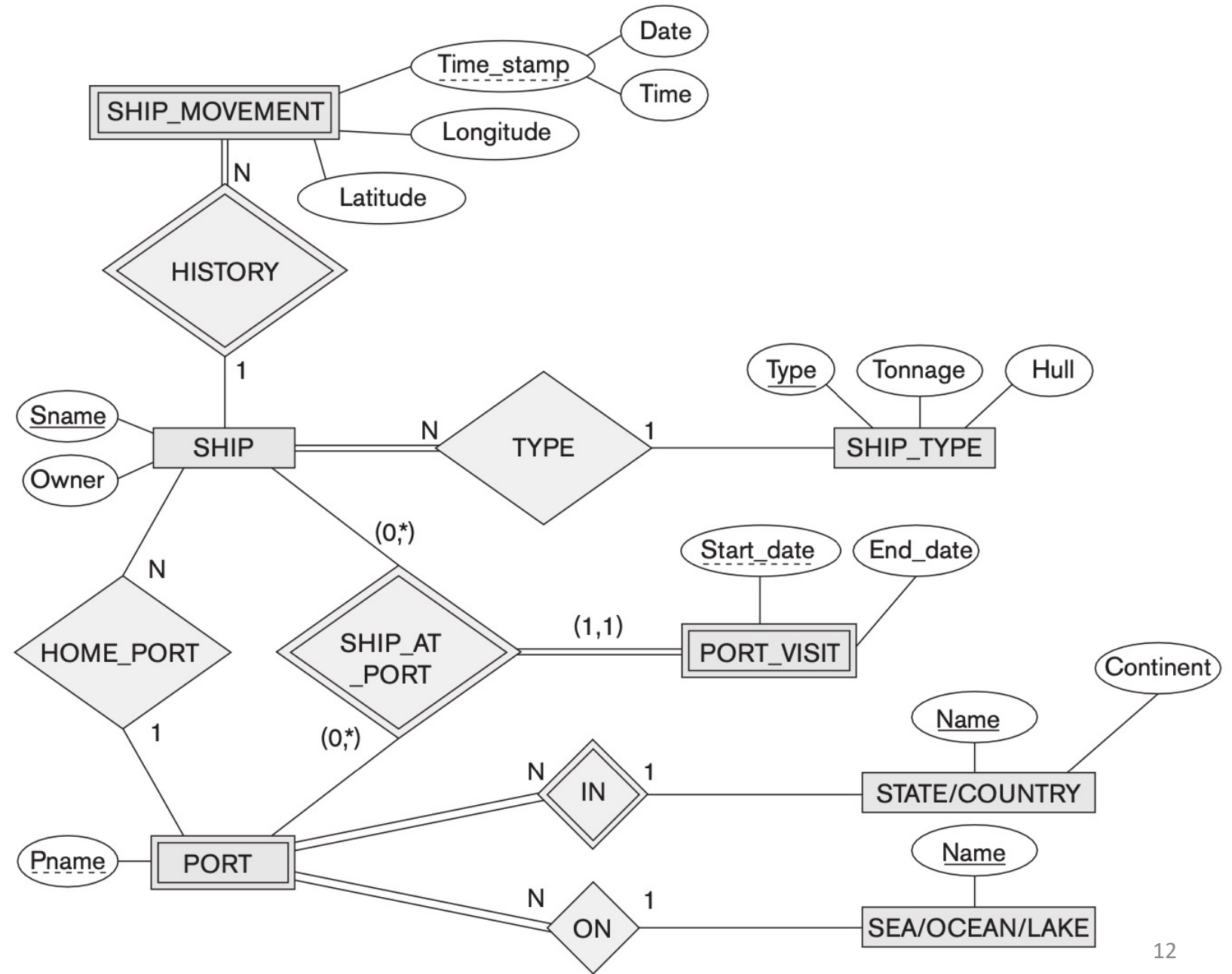
(c)



Consider n -ary relationship types where $n \geq 3$



When to
decide
superclass/
subclasses



When to decide superclass/subclasses and relationship types

When to decide superclass/subclasses and union

Connection traps

- Some people consider that problems called **connection traps** may arise with the Entity-Relationship model when designing a conceptual database schema. Two main types of connection traps are called **fan traps** and **chasm traps**. What are fan traps and chasm traps? Give examples to illustrate them. Are they really the problems with the Entity-Relationship model? If not, explain your answer and then fix the problems faced in your previously given examples.

Connection traps

□ Fan Trap

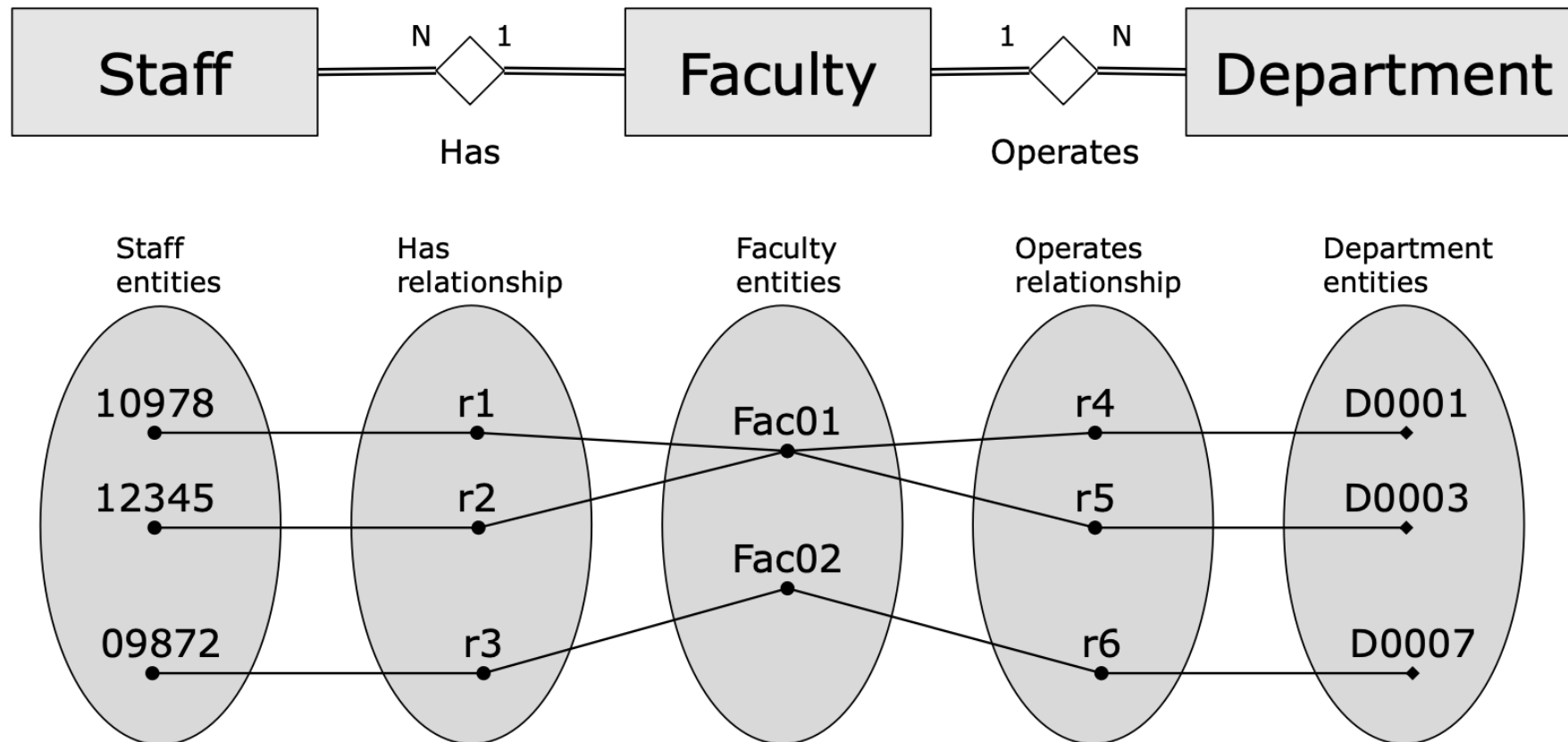
- Where a diagram represents a relationship between entity types, but pathway between certain entity occurrences is ambiguous
- Usually: two or more 1:N (M:N) relationships fan out from the same entity

□ Chasm Trap

- Where a diagram suggests the existence of a relationship between entity types, but pathway does not exist between certain entity occurrences
- Usually: optional participation

Connection traps

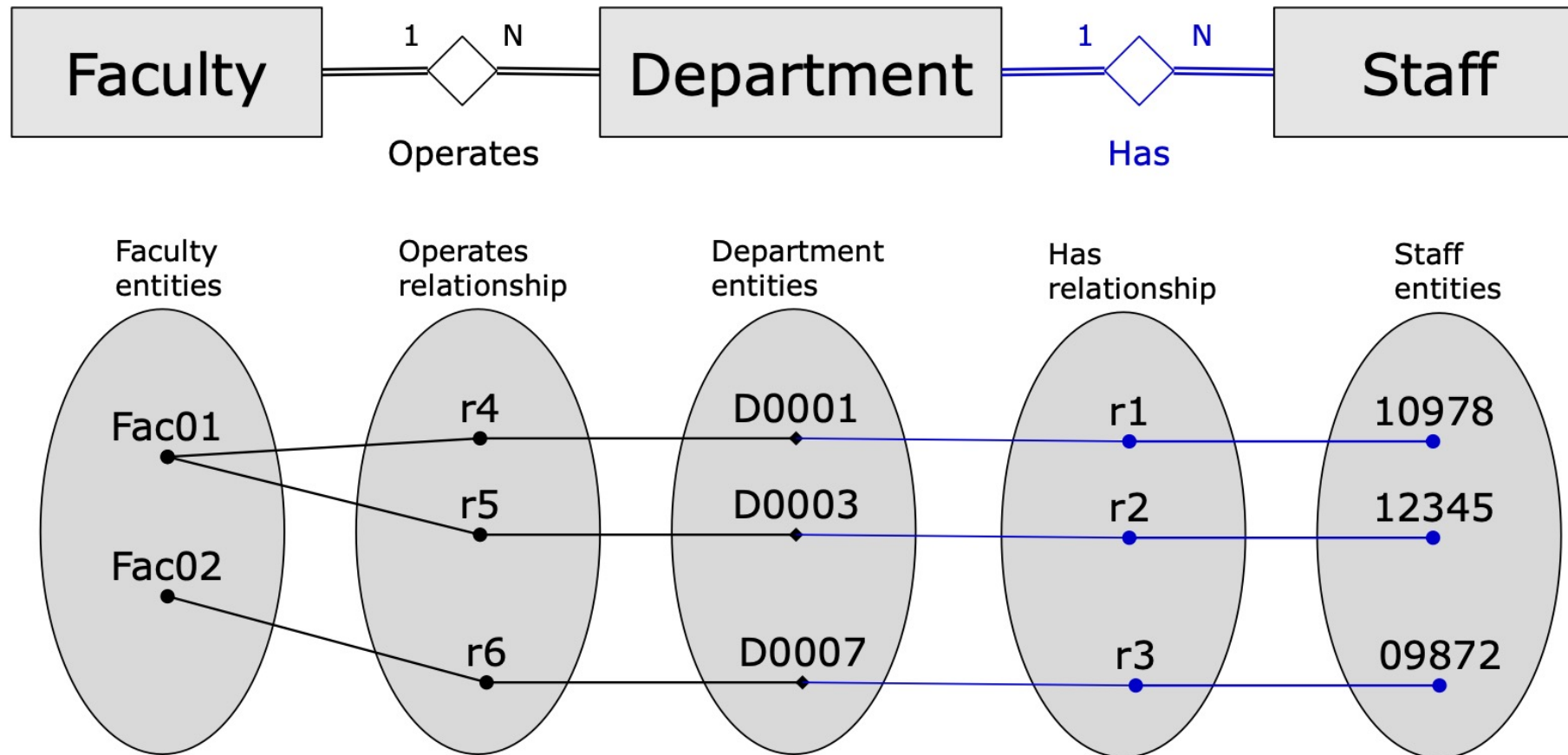
An Example of a Fan Trap



At which department does staff 12345 work?

Connection traps

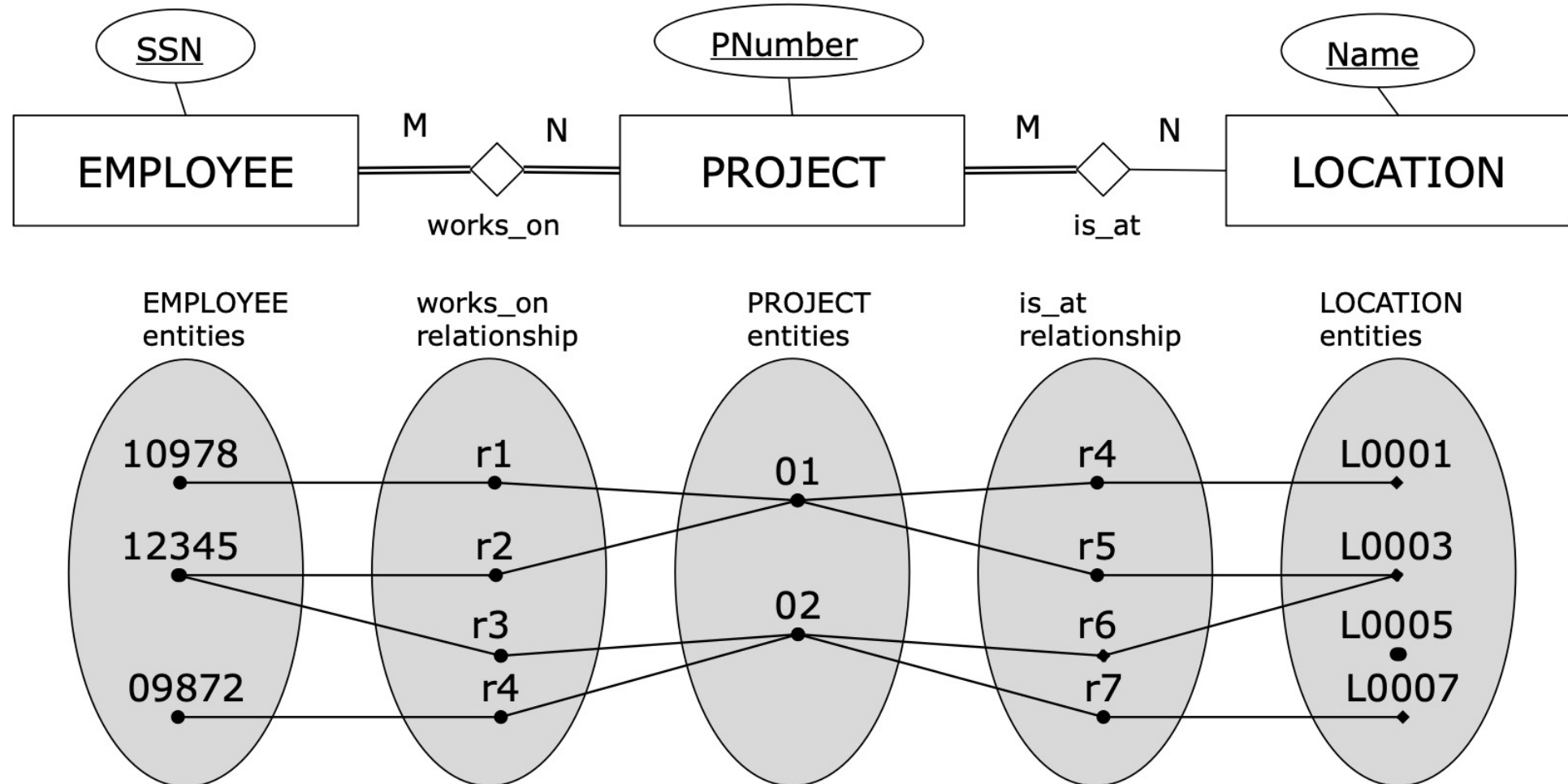
Restructured to remove the Fan Trap



12345 works at department D0003.

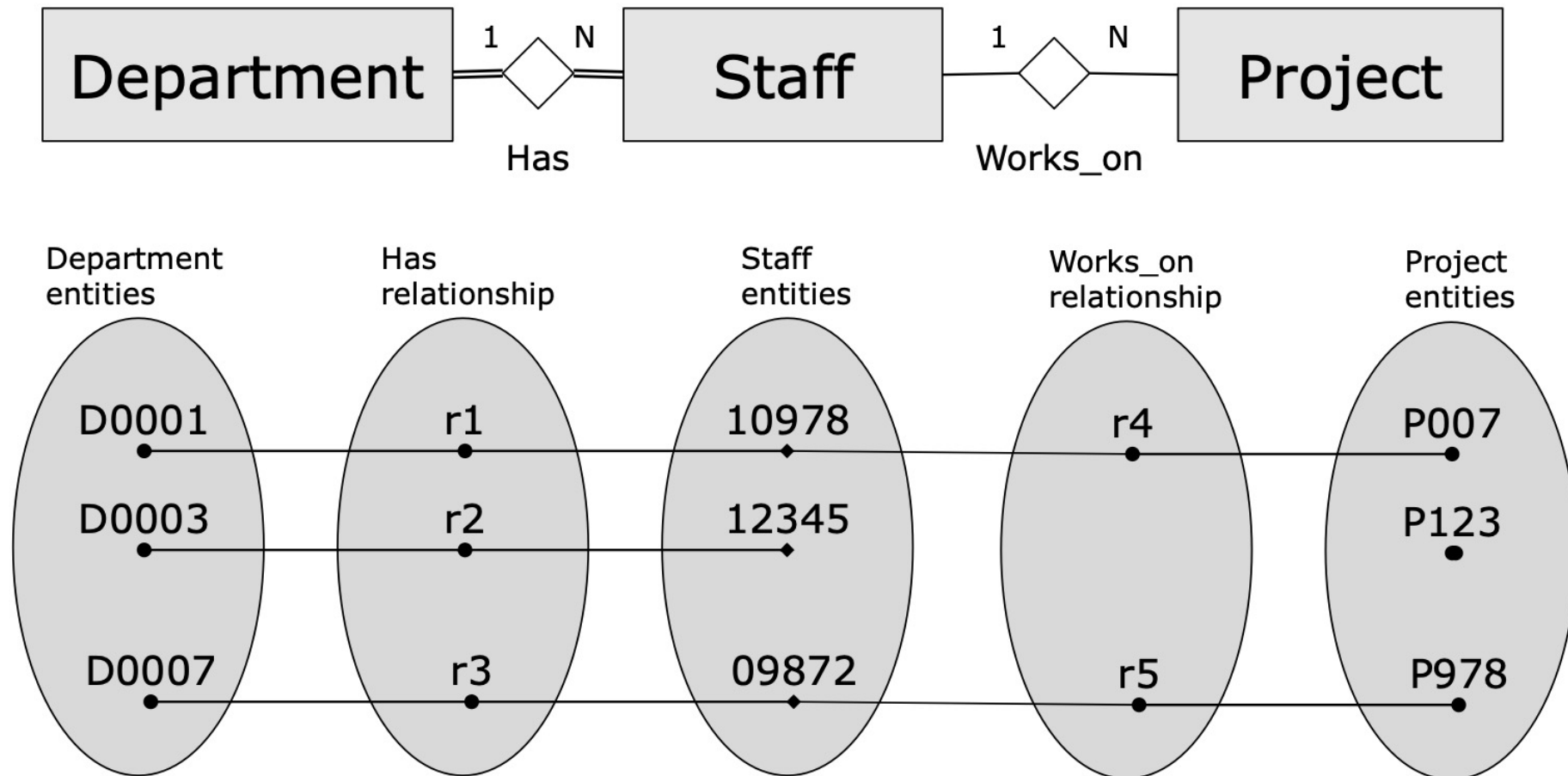
Connection traps

An Example of a Fan Trap



Connection traps

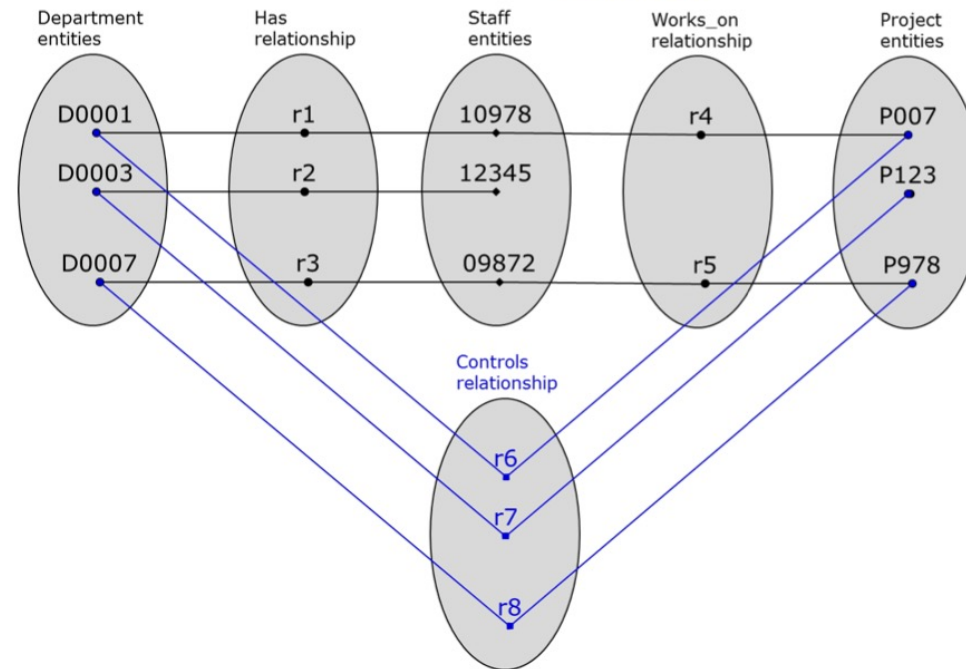
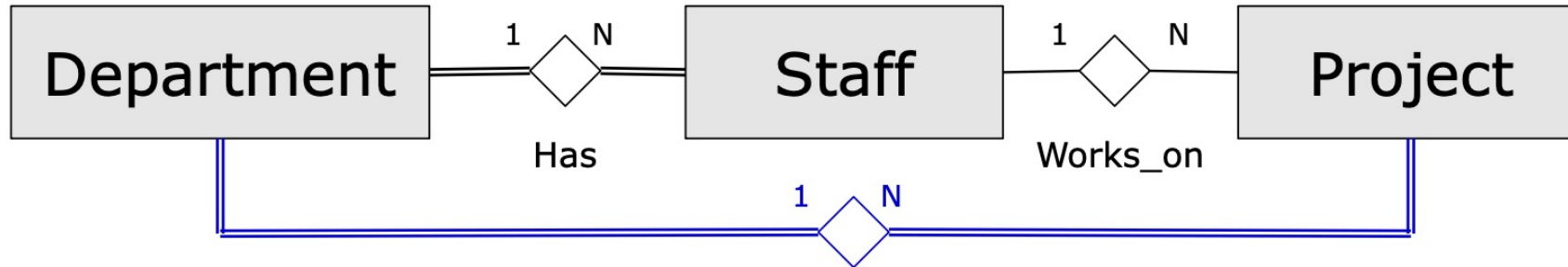
An Example of a Chasm Trap



Which department does project P123 belong to?

Connection traps

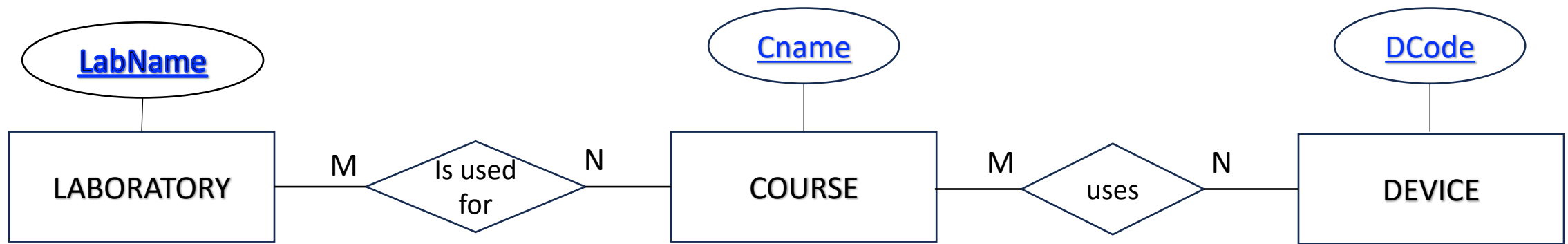
Restructured to remove the Chasm Trap



**Project P123
belongs to
department
D0003.**

Connection traps

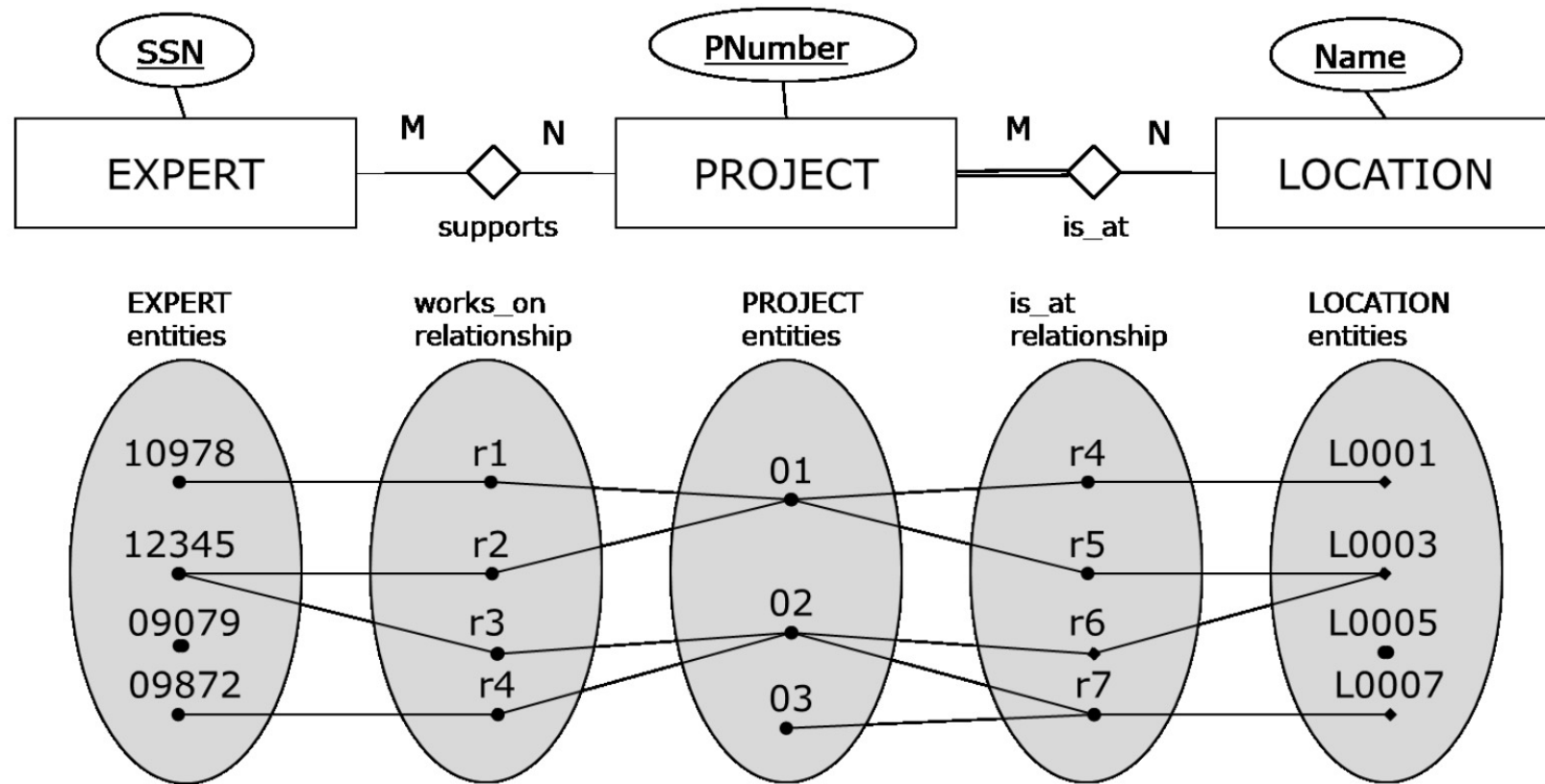
- Hybrid trap: fan trap + chasm trap



Which particular laboratory does a device belong to?

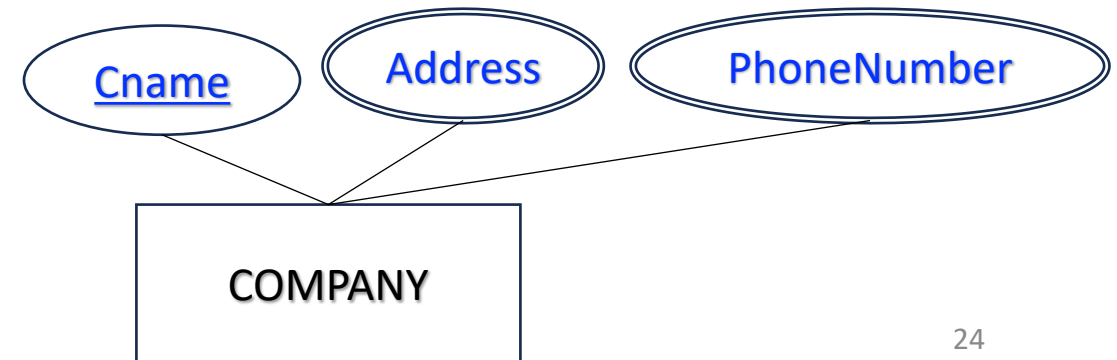
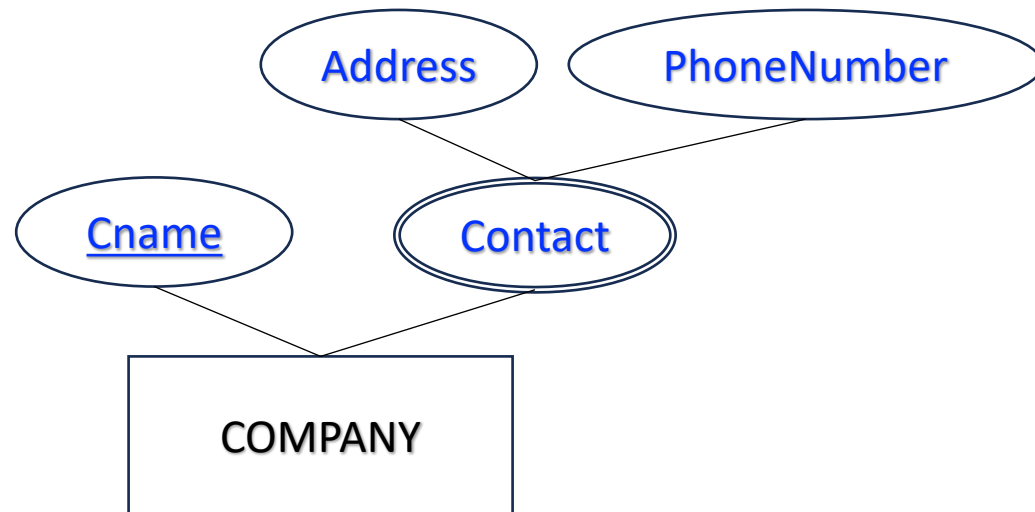
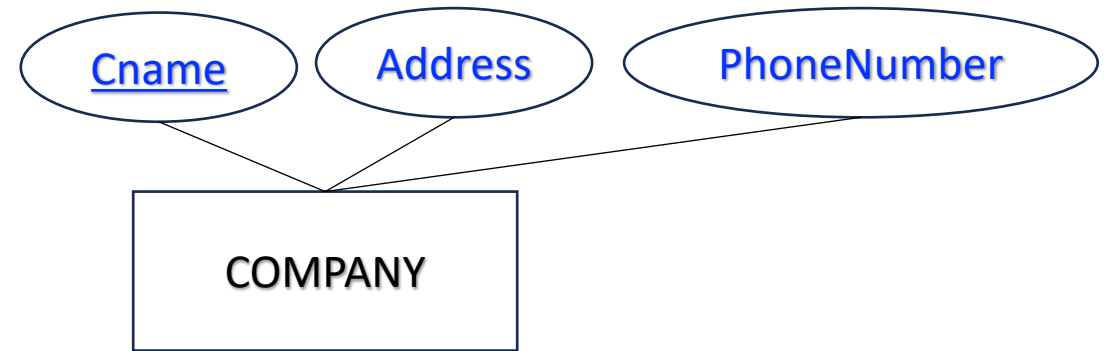
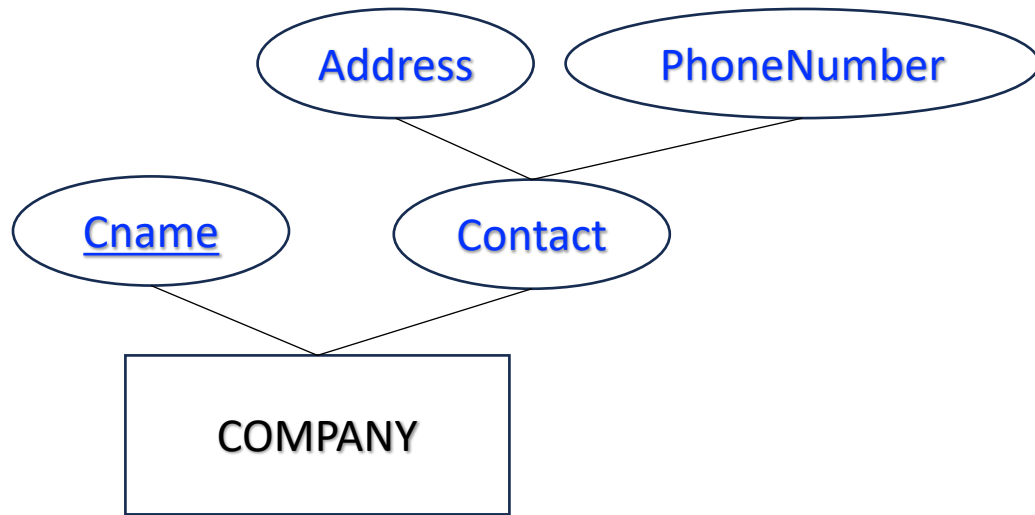
Connection traps

- Hybrid trap: fan trap + chasm trap

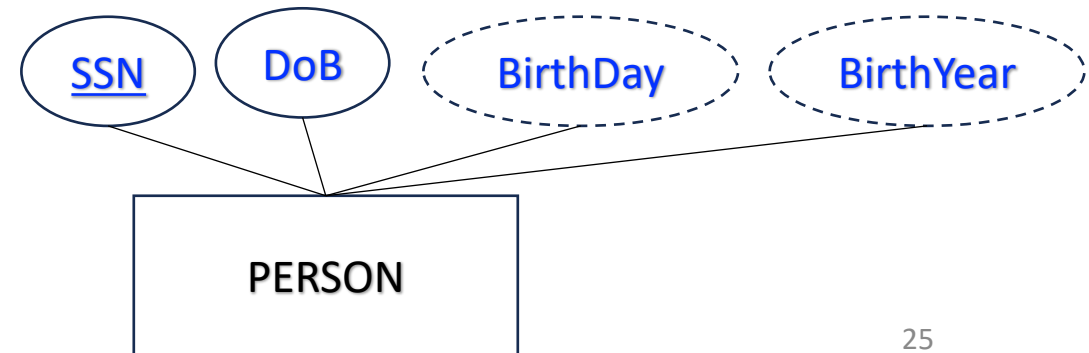
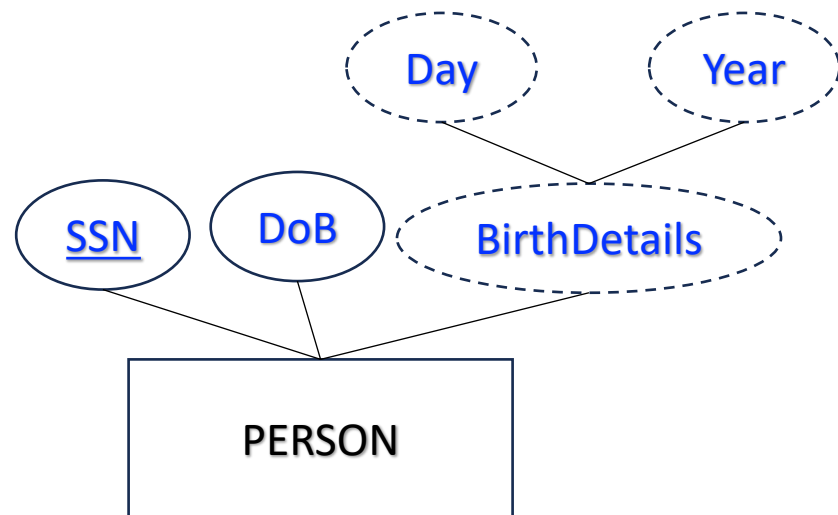
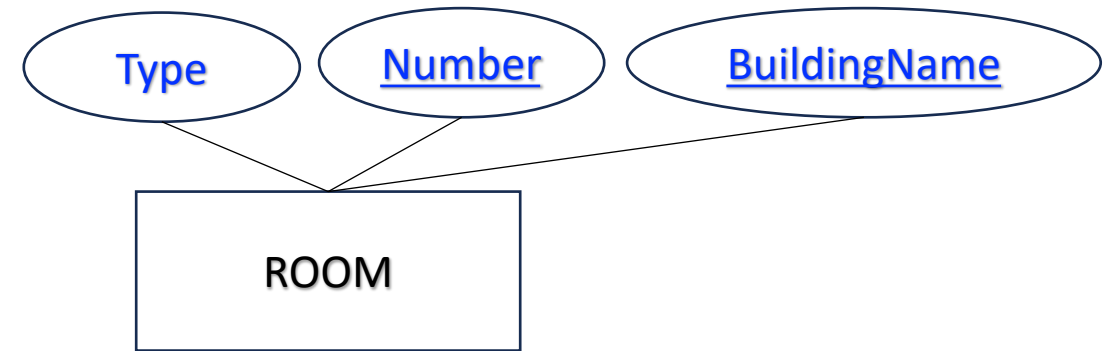
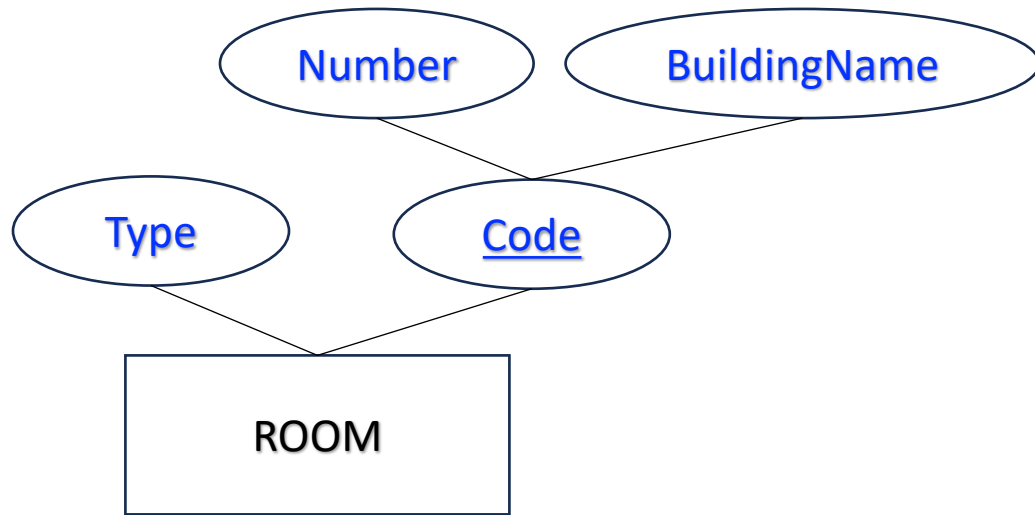


Which particular location does an expert work at?

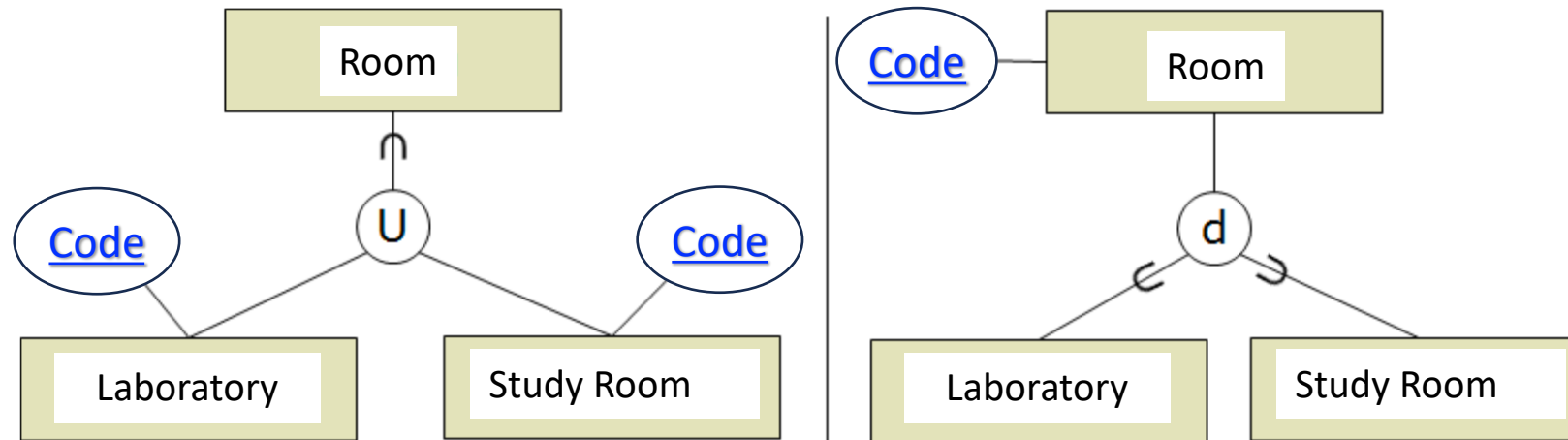
Equivalence cases - checking



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Equivalence cases - checking

